



MEMBER

Q1 2021

Ten Mistakes to Avoid

In Data Visualization

By Stan Pugsley

Ten Mistakes to Avoid

In Data Visualization

By Stan Pugsley

FOREWORD

Do any of the following sound familiar?

- You brought in reporting software but the quality of your reports has not improved
- You have a great selection of reports available but few people are using them
- Your business users prefer to build their own reports in Excel rather than use those created by the IT department
- Your business users still feel dissatisfied by reports despite projects to implement their requirements

All of these are symptoms that your reporting strategy isn't working. An advanced reporting strategy needs to walk users down a path from viewing the high-level data to getting the details they need to fixing issues or improving processes. The reports need to tell a story of what is happening, what must be improved, and how to make that improvement. Visualizations are linked in a chain leading from understanding the current state of your business to taking action to improve it.

Unfortunately, most organizations have dead-end reporting; a report may show an insight but stops at that point. There is no logical progression from that insight towards the deeper understanding and action needed to create a positive business outcome.

Following is both a framework to understand the components of a data reporting strategy and a list of mistakes that interrupt the path from data to business ROI.

© 2021 by TDWI, a division of 1105 Media, Inc. All rights reserved. Reproductions in whole or in part are prohibited except by written permission. Email requests or feedback to info@tdwi.org.

Product and company names mentioned herein may be trademarks and/or registered trademarks of their respective companies. Inclusion of a vendor, product, or service in TDWI research does not constitute an endorsement by TDWI or its management. Sponsorship of a publication should not be construed as an endorsement of the sponsor organization or validation of its claims.

MISTAKE ONE:

YOUR KPI COLLECTION IS BASED ON LEGACY REPORTS, NOT EXPERT SELECTION

Your main key performance indicator (KPI) collection must be carefully curated to align with your top business priorities. You should have a home page for each department and the real estate on that home page should be carefully guarded and organized. If reporting projects are IT-driven rather than business-driven, that real estate is often poorly used, forcing users to search for the content they wish to see.

Here are some questions to ask as you review your top KPIs:

- When you login to your reporting suite, what is the first set of KPIs or visualizations you see?
- Who decided which KPIs should be included?
- Are some of these visualizations included only because they were in a legacy report?
- How many visualizations are present in that first view?

Attention and time are precious. Make sure that when someone logs into one of your reporting tools, they see the more important items for their job function on the first page, with the KPIs aligned to the business objectives of that department.

Let's use a case study of a toilet paper manufacturer to illustrate these concepts. For our example, imagine that the company is analyzing its data as pandemic news pushed waves of supply and demand imbalances through the market. What KPIs would business users need to manage this situation?

Their top KPIs would reflect their business of making and shipping toilet paper to fill demand as efficiently as possible. The home page of their Operations reporting suite would include, all on the first page, summary metrics on:

- The sales pipeline (new demand)
- The distribution network (existing demand)
- The top manufacturing processes (supply)

If they are working on a specific initiative this year, they would highlight metrics for that initiative in a prominent place.

Action item: Create a carefully curated home page for each department to get their data visualizations. Make it a place to see, at a glance, the most important metrics for their job.

MISTAKE TWO:

KPIS AREN'T PRESENTED IN A CONCISE, LOGICAL WAY

When a KPI is displayed, it needs to be presented so that the point is clear. Too often, the important numbers are hidden between tables, stars, arrows, and bright colors and get missed. The KPIs that matter at business meetings should not have a chance of being missed, misunderstood, or confused with the other information presented.

The design of the page housing the KPIs and their charts should be dictated by what the users need to see and know. Their experience in viewing KPIs should be constructive and instructive. If the KPIs are presented in a way that makes sense and in a way that does not take extra time to mentally process or analyze, users will lean on the reporting tool rather than avoiding it in favor of shadow-IT Excel reporting.

It is also important to determine how much detail is necessary for a given report to do its job. Excessive detail does nothing but slow users down. Looking at your charts and pages through the perspective of a user is key. What would they understand? What do they want to see immediately? Are these things standing out on your page?

In our toilet paper case study, we could present the top KPIs in a 2x6 grid dashboard, with new demand on the left and supply on the right. On the top row could be current unit volumes for the product categories; on the lower row could be the time trend over the last 12 months.

Action item: Cut down the amount of distraction and unnecessary detail on the first dashboard pages. Present the needed KPIs in a well-organized, logical manner.

MISTAKE THREE:

USERS ARE UNABLE TO TELL WHAT CHANGES ARE SIGNIFICANT

Significant shifts need to be highlighted and obvious to those viewing your charts. A chart should not be presented as a subjective interpretation by any individual but instead should be designed to highlight changes based on deviation from a historical average. If there is a change in the data that affects current operations, whether beneficial or harmful, that change should stand out from the other information.

Often, a chart can be too busy for anyone to pick out the important changes and may even lead users to focus on unimportant details. Although many changes may be occurring at once, not all of these will be important to evaluate and possibly act on. Focus on those changes important to operations and highlight the facts using color, symbols, or bolding. Meanwhile, let the trivial changes sit in the background so as not to detract from the important matters at hand.

Finding the balance between simplicity and emphasis is key. A chart without any emphasis leads to wasted time spent looking for changes that should have stood out to users. However, a dashboard with too many attention-drawing colors and points of emphasis can become cluttered and distracting. If you are not sure what is statistically significant, a 95 percent confidence limit is usually a good starting point.

Back to our toilet paper company. If a sudden spike in sales occurs in a given region, put this situation in red on your line chart or make text in a table boldface to draw the user's attention. If properly emphasized, users can take appropriate and timely actions. Instead of spending too much time and effort squinting at the chart to see whether any change has occurred, the change will be obvious to the business user and allow for that time to be better spent making plans and taking action to respond to that change or proactively prepare for any future change the data suggests is coming.

Action item: Make the significant changes stand out on your chart so problems are apparent to any user, not just those highly familiar with the numbers or processes.

MISTAKE FOUR:

AN INCORRECT SELECTION OF CHART TYPES HIDES YOUR MESSAGE

Each type of chart is built to tell a particular story. Therefore, the decision about which type of chart to use in your business environment is not simply a matter of design and stylistic choice. Your chart needs to serve a function.

For example, charts can be specialized in comparing categories, showing change over time, visually representing information, or connecting groups. Understanding the purpose and function of each chart type is key. If these purposes are not understood, the chart chosen may send the wrong message or hide the important details completely. This will impede and reverse progress for your users.

For example, using a bar chart when users need to notice a change over time will lead to confusion and a missed message because bar charts are simply not built to serve this function. It cannot be overstated that choosing a chart is not a matter of graphic design. Charts are functional, not just fun things to look at.

The TDWI Upside article “Data Story Visualization: A Decision Tree” (available at <https://tdwi.org/articles/2020/10/06/bi-all-data-story-visualization-decision-tree.aspx>) presents a useful decision tree that can help you make the right visualization choice.

In the toilet paper company, we would expect to see line charts showing the time trend of supply and demand. Then we could use large scorecard or bullet charts to highlight the percent change month-over-month (or whatever the appropriate time period is).

Action item: Understanding what each chart type displays helps make sure your chart choice matches the story you want to tell.

MISTAKE FIVE:

KPIS ARE NOT PAIRED WITH DETAILED REPORTS AND DRILL THROUGH

Charts can show an insight, but they also need to be built for investigation. When a chart simply shows the summary of a situation in the company without the option for more detailed reports or ways to drill down into the KPIs, it becomes a dead end. You cannot predict what the next step for an analyst may be when they reach a dead end.

KPIs should always be partnered with drill-down paths to their details and background information to promote greater understanding. Making these details available at the discretion of the users is important because they will be using the reporting tools in a world that changes every day. New details will be necessary for different occasions, so making them available is key.

Think of your chart as a gateway to learning more and gaining a greater understanding of what is actually going on with the KPIs. Do not let the information stop at “there was an increase in sales.” Let it continue with “by how much?” and “what does this mean now?”

As extra time is put in to make more information accessible through your drill-down-enabled charts, you will empower a broader group of business analysts and managers to explore the data, creating a better foundation for a data-driven organization because analysis becomes easier to perform.

In our toilet paper example, a line chart showing a change in sales should be accompanied by a drill-down path to see

- Which SKUs are low?
- Is the shortage similar in all regions?
- Is the shortage due to low supply or high demand?

Each KPI should be a gateway to an analysis path. A chart showing demand by product category would be an entry point to a drill-down to analyze the numbers by SKU. A chart showing total sales per region should be just a starting point to drill-down to a full geographic analysis by, say, state or city.

Action item: Complete your charts on KPIs with features allowing users to drill down and obtain more information about the numbers that matter.

MISTAKE SIX:

BUSINESS USERS CANNOT PERFORM AD HOC ANALYSIS

Have you empowered your business community to do their own ad hoc analysis rather than depend on a small group of expert analysts? Do you have a well-organized data warehouse that makes self-service reporting feasible?

Making data well-organized and available and allowing for business users to do ad hoc analysis allows these users to respond quickly to business changes. Every day presents a new problem and a new question for business users. If they can see the data at a different angle in a new chart by a simple click, they can better understand their business situations. They will be inclined to turn to your reporting tools to solve these problems.

Users may have new requirements every time they approach data. IT projects will never satisfy every future business requirement. Rather, IT teams should focus on deploying ad hoc analysis environments that allow self-service for any future requirements. Business users should not have to turn to you to make things work. Your design can allow them to work things out on their own.

At the toilet paper company, it's unlikely the IT department anticipated the business requirements for reporting during a pandemic. If they have a well-built, self-service reporting environment, multiple users from multiple departments could discover and report on the emerging trends using consistent numbers. Business users would be able to see the shock to their supply chain and perform analysis without ever contacting the IT department for help.

Action item: Focus efforts on building a robust self-service analysis environment rather than trying to capture and meet exhaustive lists of business requirements in canned reports.

MISTAKE SEVEN:

LOGICAL NEXT STEPS HAVE NOT BEEN AGREED UPON

Once a problem is identified by a chart, logical next steps should be laid out within your tools. Create paths for analysis and resolution. You can do this by linking KPIs in a chain that leads the users down the logical paths of analysis and resolution. For example, if inventory is low, the logical next step would be to review purchase orders and shipment information. Those business processes are all connected, so their KPIs should also be connected.

If the toilet paper inventory is low, your reporting should lead your users down the alternate paths to replenish inventory:

- Option1: Look for alternate SKUs to fill the needs
- Option2: Check the production lead time to meet demand
- Option3: Ship product(s) between warehouses if imbalances exist

Strategic reporting means working with the business to understand the dependencies in business processes and reflect the linkages in your reporting.

Action item: Create links in each dashboard or KPI to one or more logical next steps for analysis or action.

MISTAKE EIGHT:

VISUALIZATIONS ARE NOT ORGANIZED TO LEAD TO ACTION

The visualizations you create should not just point out a problem. They should be designed to lead to action. If we build visualizations based only on a list of business requirements, those stakeholders defining the report requirements will assume that the action is known to future users. They give requirements assuming the users will be themselves!

A strategic visualization chain should walk users down a path from initial insight to final action. It should not assume that users have long institutional knowledge to know intuitively what to do with the insights from data.

In our case study, we could create an inventory dashboard specifically designed to show the alternatives to deal with excess or low inventory. The KPIs on the dashboard would show the relationship between supply and demand and give information needed to address the imbalances.

Action item: Review your dashboard to identify any dead-end points not linked to related business actions.

MISTAKE NINE:

LACK OF LINKS TO RELATED APPLICATIONS

The best reporting tools will lead users to solutions to their problems. If links are given to related applications, programs, and sites that will make solutions attainable, action can be taken in the moment. Instead of just leaving users with an idea of what to do next or—even worse—with simply the idea that a problem exists, make their path easier by giving them a direct link to solutions.

Sending users to the next step could include providing links to supplier portals, financial applications, shipper websites with tracking numbers, or contact information for people who could be helpful in their situation.

Try placing useful links to related applications on the right of your visualization. Eyes of viewers will naturally flow in that direction after taking in information from the visualization itself. A hover-over feature may also be helpful if connected to items on a chart that show extremes. There could be links connected with hovering over a point that help in these extreme cases.

With the toilet paper company, the reporting solution should, at the proper locations:

- Embed links to a production management system (order more)
- Embed links to a distribution management system (ship more)
- Embed links to a pricing tool (raise prices)

Action item: Provide links to applications and people that help your users solve their problems. Embed these links directly in your visualization.

MISTAKE TEN:

PASSWORD AND SECURITY OBSTACLES HINDER USERS FROM TAKING ACTION

The momentum is built up and your users are now using the connecting links you have provided so they can take action. However, password and security obstacles will kill this momentum. When users are required to dig out passwords they never remember making (or passwords they did not even make themselves) or reach out to other departments seeking access to their solutions because of security obstacles, they will get frustrated and distracted.

Single-sign-on (SSO) integration of your applications, including the reporting tools, is a key step to connecting data to business outcomes. Ask business users to document and report on the obstacles they face as they switch between applications on the path from data to their final action. Create a backlog of projects to resolve those obstacles.

Action item: Ensure smooth and easy access between steps from reporting to the business process by removing unnecessary password and security obstacles.

ABOUT THE AUTHOR

Stan Pugsley is a data warehouse and analytics consultant with [Eide Bailly Technology Consulting](#) based in Salt Lake City, UT. He is also an adjunct faculty member at the University of Utah Eccles School of Business. You can reach the author via email (SPugsley@EideBailly.com) or visit his LinkedIn page: <https://www.linkedin.com/in/spugsley/>. The author was assisted by his daughter Meg Pugsley, an undergraduate student at Brigham Young University, in the preparation of this material.

ABOUT **TDWI**

TDWI is your source for in-depth education and research on all things data. For 25 years, TDWI has been helping data professionals get smarter so the companies they work for can innovate and grow faster. TDWI provides individuals and teams with a comprehensive portfolio of business and technical education and research to acquire the knowledge and skills they need, when and where they need them. The in-depth, best-practices-based information TDWI offers can be quickly applied to develop world-class talent across your organization's business and IT functions to enhance analytical, data-driven decision making and performance. TDWI advances the art and science of realizing business value from data by providing an objective forum where industry experts, solution providers, and practitioners can explore and enhance data competencies, practices, and technologies. TDWI offers conferences, topical seminars, onsite education, a worldwide membership program, business intelligence certification, live webinars, resourceful publications, industry news, an in-depth research program, and a comprehensive website: tdwi.org.



**Transforming Data
With Intelligence™**

555 S. Renton Village Place, Ste. 700
Renton, WA 98057-3295

T 425.277.9126
F 425.687.2842
E info@tdwi.org
tdwi.org